

```
public class BinarySearch {
    public static int binarySearch(int[] arr, int target) {
        int left = 0;
        int right = arr.length - 1;
        while (left <= right) {
            int mid = left + (right - left) / 2;
            if (arr[mid] == target) {
                return mid;
            }
            else if (arr[mid] < target) {
                left = mid + 1;
            }
            else {
                right = mid - 1;
            }
        }
        return -1;
    }
    public static void main(String[] args) {
        int[] arr = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10};
        int target = 7;
        int result = binarySearch(arr, target);
        System.out.println(result != -1 ? "Элемент " + target + " найден на индексе: " + result : "Элемент не найден.");
    }
}
```